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A facile hydrothermal approach towards photoluminescent carbon dots from amino acids

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ABSTRACT

A facile one-pot method to fabricate photoluminescent carbon dots (CDs) was developed by the hydrothermal treatment of amino acids at mild temperatures. Derived from three different kinds of amino acids including serine, histidine, and cystine, the resultant CDs show uniform spherical morphology with the diameters in the range of ~2.5–4.7 nm. These amino acid derived CDs also manifest excellent photoluminescence behavior with the quantum yields (QYs) of ~7.5% and high stability. More importantly, this method provides the opportunity to modify the sizes, structures, and photoluminescent behavior of CDs by the utilization of diversified amino acids with different structural characteristics.

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1. Introduction

In the last few years, photoluminescent carbon dots (CDs) have attracted enormous attention due to their fascinating virtues including stable photophysical properties, good dispensability in aqueous solution, high resistance to photo-bleaching, environmental security, low cytotoxicity, excellent biocompatibility and so on, which make them appealing fluorescent materials in photocatalysis, energy conversion, optoelectronics, biological labeling, cellular imaging, and drug delivery [1–12]. According to the different starting materials, the fabrication strategies of CDs can be categorized as top-down and bottom-up approaches. Generally, the segmentation of bulky carbon-rich precursors is named as top-down methods [13–18]. Correspondingly, the bottom-up methods refer to the thermal or solvothermal conversion of small molecules or polymers to CDs [19–24]. Among various bottom-up methods, hydrothermal treatment of the precursors provides a mild, efficient and broadly applicable route to CDs [25]. Moreover, the structural and photo-physical properties of the CDs derived from hydrothermal treatment are greatly decided by the starting materials. Therefore, the delicate selection of precursors for the hydrothermal production of CDs is of great importance in the exploration of unprecedented CDs with excellent photoluminescent behavior.

As the basic building units of protein, amino acids are abundant, inexpensive and biocompatible. They possess both amino and carboxyl groups, which enable them to be easily cross-linked via amide groups. Moreover, it has been reported that the size, morphology, crystalline degrees and composition of the CDs could greatly affect their photoluminescent behavior [21,26]. The rich content of heteroatoms such as nitrogen (N) and sulfur (S) in amino acids will render the resulting CDs to contain heteroatoms in the carbon framework, which is expected to influence their photoluminescent properties accordingly. These advantages thus make amino acids the ideal precursors for the bottom-up construction of CDs.

Herein, we report a facile one-pot approach towards photoluminescent CDs by the hydrothermal treatment of different amino acids including serine (Ser), histidine (His), and cystine (Cys) at mild temperatures (Scheme 1). As the results, CDs with uniformly spherical morphology and narrow size distribution were obtained. With the diameters ranging from ~2.5–4.7 nm, the resultant CDs manifest the quantum yields (QYs) of ~7.5% and high stability, which are comparable to those of the CDs obtained at high temperatures over 900 °C [2,5,21,26].

2. Experimental section

2.1. Materials

L-Serine and L-Histidine were purchased from Shanghai Aladdin Chemical Reagent Company. L-Cystine and acetic acid were purchased from Sinopharm Chemical Reagent Company. All

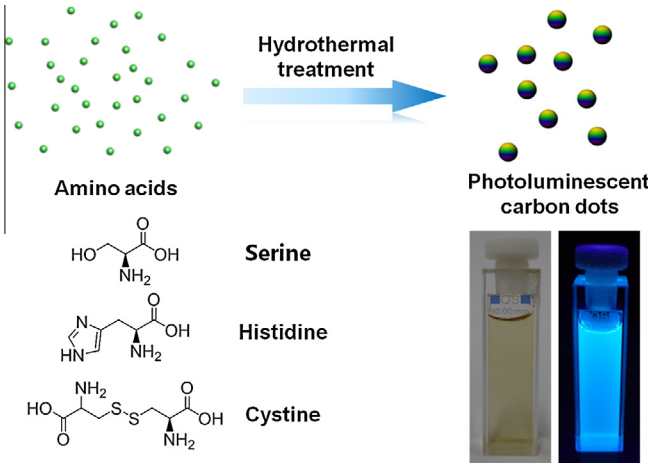
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chemicals were used as received without any further purification. Ultrapure water (18.2 MΩ cm @25 °C) was used in all experiments.

2.2. Synthesis of the CDs

In a typical synthesis procedure, 0.0125 mol amino acids were first dissolved in 36 ml acetic acid (2 wt%). The mixtures were stirred at room temperature for several minutes and sealed in a 50 ml stainless steel autoclave for the following hydrothermal treatment for 12 h at 200, 220 and 250 °C, respectively. It should be noted that the amino acids only had good solubility in acidic solutions and the utilization of neutral or basic solution would lead to the precipitation of the precursors. Therefore, diluted acetic acid was used as the solvent for the hydrothermal synthesis of CDs. After the hydrothermal process, the color of the mixture solution turned from colorless into dark yellow, implying the successful conversion of amino acids to carbon nanomaterials [25]. The obtained suspension was then centrifuged at a 10,000 rpm for 10 min and the supernatants were dialyzed against Milli-Q water with a cellulose ester membrane bag (Mw = 3500) for 24 h to remove the excess precursors. After filtering through 0.2 μm Teflon filter, a clear, light yellow aqueous suspension was finally obtained. According to the carbon source and the temperatures of hydrothermal treatment, the resultant CDs are abbreviated as CDs–*X*–*T*, where *X* refers to the amino acid (Ser, His, and Cys), and *T* stands for the hydrothermal temperatures (200, 220 and 250).



Scheme 1. A schematic illustration for the preparation procedure of CDs by hydrothermal treatment of amino acids. Inset: photo image of the solutions of CDs-His-200. Left: under daylight; right: under 365 nm UV irradiation.

2.3. Characterizations

TEM measurements were performed on JEM-2010F at operating voltage of 200 kV. The sample was diluted in ultrapure water and the suspension dropped on carbon-coated copper grid by evaporation in air. UV/Vis spectra were recorded at room temperature on a Hitachi J-4100 spectrophotometer. Fluorescence spectra were recorded for progressively longer excitation wavelengths from 300 to 480 nm in 20 nm increments on a Horiba Fluoromax-4 spectrometer.

3. Results and discussion

The morphology and microstructure of the CDs were first investigated by transition electron microscopy (TEM). As indicated in Figs. 1, S1 and S2, all the CDs exhibit similar spherical structures. For the CDs from serine, the diameters of CDs-Ser-200,

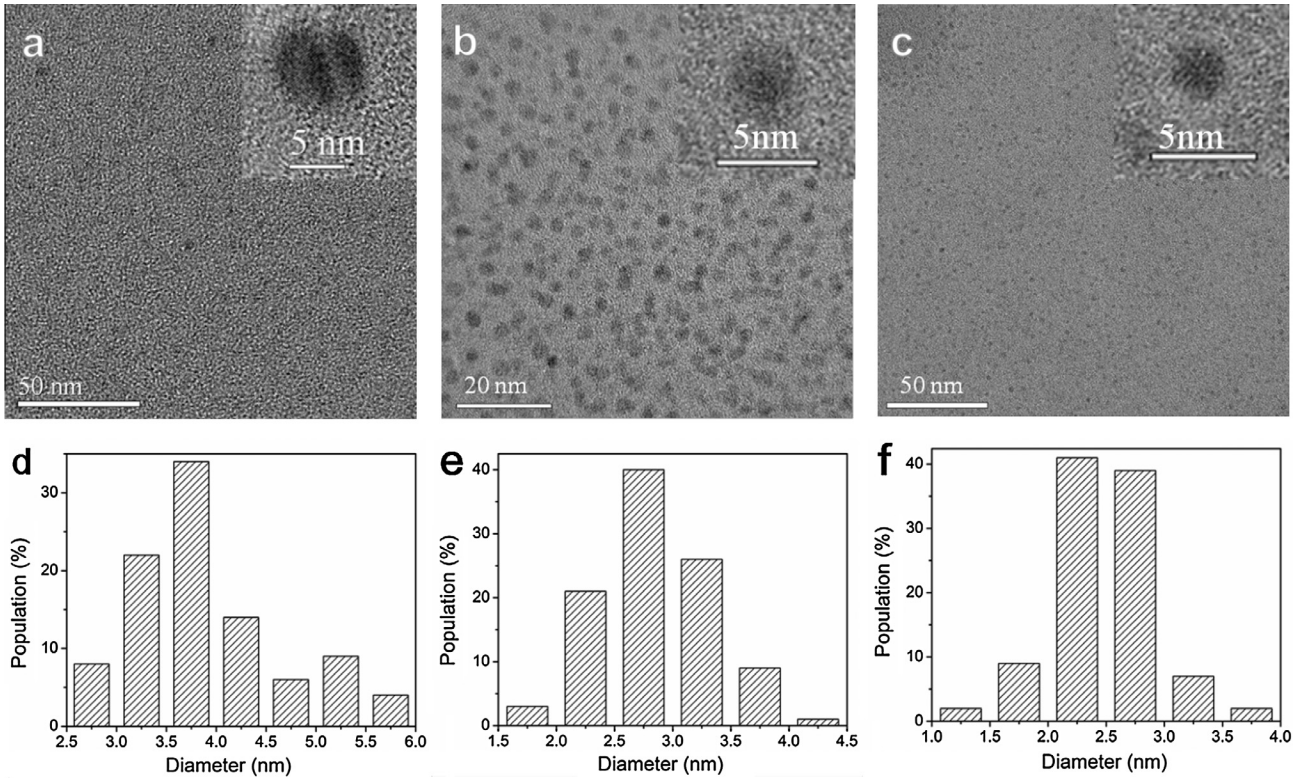
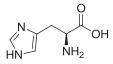
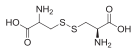


Fig. 1. TEM images of (a) CDs-Ser-200, (b) CDs-Ser-220 and (c) CDs-Ser-250. Inset: the corresponding HRTEM images of the CDs; the particles size distribution histograms of (d) CDs-Ser-200, (e) CDs-Ser-220 and (f) CDs-Ser-250 obtained from 100 particles.

Table 1
The structural and photophysical properties of the CDs from amino acids.

Amino acid	Structural formula	Sample name	Particles size (nm)	Adsorption peak (nm)	Max. PL peak (nm)	QY (%)
Ser		CDs-Ser-200	3.7 ± 2.0	261	418[a]	5.1
		CDs-Ser-220	2.7 ± 1.5	261	418[a]	3.5
		CDs-Ser-250	2.5 ± 1.5	261	413[a]	2.2
His		CDs-His-200	2.7 ± 1.5	296	416[a]	5.5
		CDs-His-220	3.2 ± 0.5	296	415[a]	3.5
		CDs-His-250	3.0 ± 1.8	272	403[a]	2.7
Cys		CDs-Cys-200	4.7 ± 3.0	270, 310	390[b]	7.5
		CDs-Cys-220	4.2 ± 2.0	270	379[b]	2.9
		CDs-Cys-250	3.2 ± 2.0	270	379[b]	2.6

[a] Excited at 340 nm.
[b] Excited at 320 nm.

CDs-Ser-220 and CDs-Ser-250 are 3.7 ± 2.0, 2.7 ± 1.5, and 2.5 ± 1.5 nm (Fig. 1), respectively, which show an obvious trend of decrease with the increasing of the hydrothermal temperatures. The similar reduction of particle sizes with the increment of the temperatures can also be found in the CDs from cystine (Fig. S2). However, for the CDs derived from histidine (Fig. S1), CDs-His-220 has a larger diameter (3.2 ± 0.5 nm) than those of CDs-His-200 (2.7 ± 1.5 nm) and CDs-His-250 (3.0 ± 1.8 nm). The different variation trends of the diameters of CDs may be attributed to the different chemical structures of their precursors. Among the three amino acids, both serine and cystine are just aliphatic amino acids, while histidine contains one heterocyclic imidazole ring, which can effectively prevent the shrinkage of the carbon framework during the thermal treatment [27–29].

On the other hand, it seems that the molecular weight of the amino acids also have some relationship with the sizes of the CDs. As summarized in Table 1, the CDs from serine, the amino acid with lowest molecular weight in the three precursors, have the smaller diameter than those from the other amino acids. On the contrary, the CDs from the largest amino acid cystine are the biggest ones among all the samples. Moreover, it is notable that the structures of the precursors also have profound influence on the graphitic degree. According to the high-resolution TEM (HRTEM) images (inset of Figs. 1 and S2), the diffraction contrasts of CDs from serine and cystine are very low and do not contain any obvious lattice fringes, indicative of their amorphous nature. In contrast, the histidine generated CDs have well-resolved lattice (Fig. S1), suggesting the high crystallinity of the carbon framework, which is also owing to the aromatic imidazole ring in histidine [21].

In order to explore the optical properties of the amino acid-generated CDs, their UV–Vis absorption and photoluminescence emission spectra were recorded accordingly. As illustrated in Fig. 2a, the CDs from serine display the similar UV–Vis absorption spectra with an absorption band at ca. 260 nm, assigning to the π–π^* transition of carbon atoms, which is similar to that of previously reported CDs derived from protein and soy milk [22,30]. With the increase of the hydrothermal temperature, the intensity of the peak shows a weak increment. In contrast, those CDs from histidine manifest much different adsorption behavior (Fig. 2b) [21]. The UV–vis absorption spectra of CDs-His-200 show a strong absorption peak centered at 296 nm, which is very close to the characteristic absorption spectra of histidine [31,32]. In contrast, CDs-His-250 gives an apparently blue-shift to 271 nm, indicative of the enhanced π–π^* transition within the graphitic framework of the CDs [18,21,30]. The adsorption of CDs-His-220 shows the intermediate between those of CDs-His-200 and CDs-His-250, and the gradually changed adsorption spectra as the hydrothermal temperatures increase imply the improved integration of all the components in the CDs. With regard to the CDs from the sulfur containing cystine, it is worth noting that

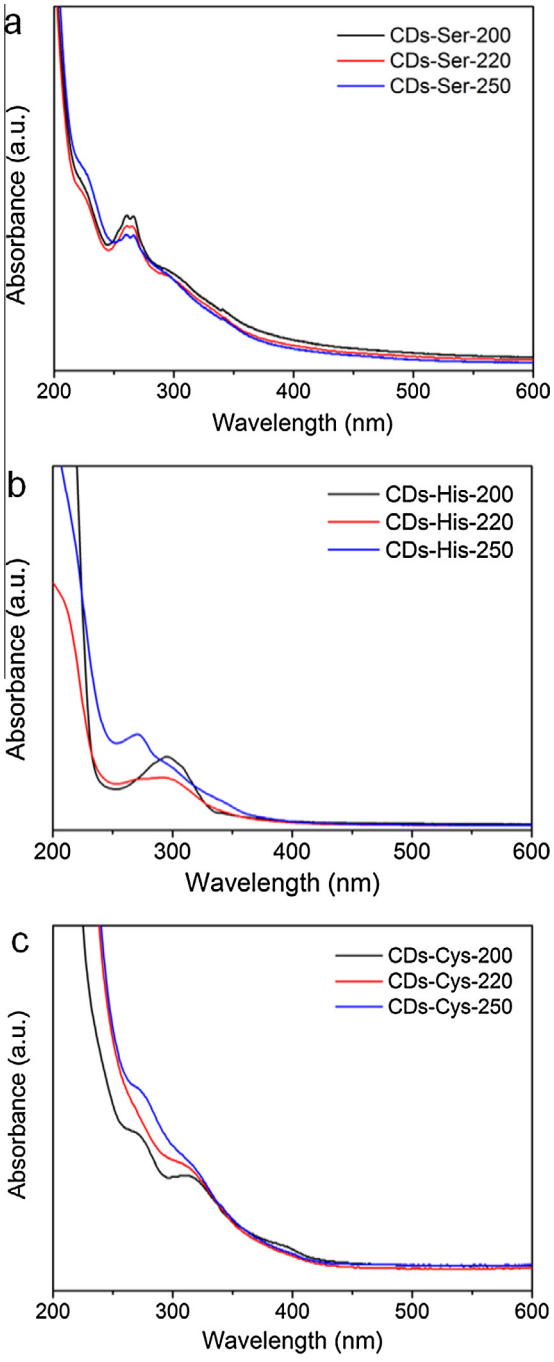


Fig. 2. UV/Vis absorption spectra of the CDs at three different temperatures (a) the CDs from serine; (b) the CDs from histidine; (c) the CDs from cystine.

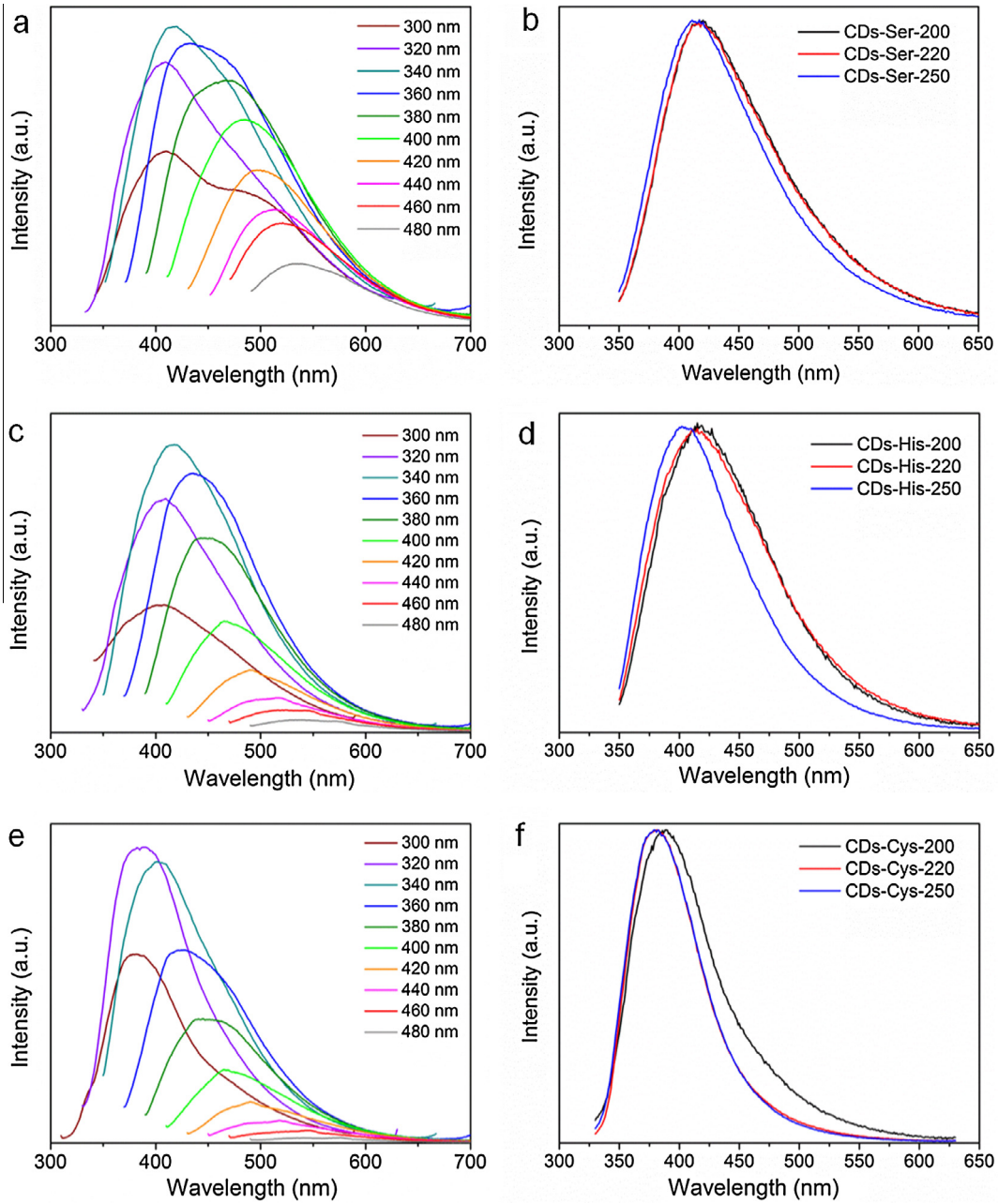


Fig. 3. (a) Photoluminescence emission spectra of CDs-Ser-200; (b) normalized emission spectra of the CDs-Ser-200, CDs-Ser-220 and CDs-Ser-250 under the maximum excitation wavelength of 340 nm. (c) photoluminescence emission spectra of CDs-His-200; (d) normalized emission spectra of the CDs-His-200, CDs-His-220 and CDs-His-250 under the excitation wavelength of 340 nm. (e) photoluminescence emission spectra of CDs-Cys-200; (f) normalized emission spectra of the CDs-Cys-200, CDs-Cys-220 and CDs-Cys-250 under the excitation wavelength of 340 nm.

the UV–Vis spectra of CDs-Cys-200 (Fig. 2c) contain two distinct absorption peaks at 310 and 270 nm, which have been also observed in the cystine derived compounds [33]. With the increased hydrothermal temperature, these two peaks are fused as one shoulder peak in the UV–Vis spectra of CDs-Cys-220 and CDs-Cys-250, suggesting the better combination of the sulfur atoms in the CDs.

The photoluminescent behaviors of the amino acid-generated CDs also demonstrate a strong dependence on their precursors. Along with the changed excitation wavelength from 300 to 480 nm, the photoluminescence peak of the CDs from serine red-shifted from 410 to 620 nm (Figs. 3a and S3). This excitation-dependent photoluminescence behavior has been extensively reported in fluorescent carbon-based nanomaterials, which might be due to the optical selection of differently sized CDs and their surface defects [34,35]. Furthermore, the normalized emission spectra of CDs-Ser-200, CDs-Ser-220 and CDs-Ser-250 all showed the same maximum photoluminescence emission at 410 nm with the excitation wavelength of 340 nm (Fig. 3b). In the case of the CDs from histidine, excitation-dependent photoluminescence behaviors similar to the CDs from serine can also be observed (Figs. 3c and S4). Along with the changed excitation wavelength from 300 to 480 nm, the emission spectra of CDs-His-200 are broad, ranging from 400 (purple) to 530 nm (green[1]). At the same excitation wavelength of 340 nm, the peaks of photoluminescence emission of CDs-His-200, CDs-His-220 and CDs-His-250 demonstrate

[1] For interpretation of color in 'Fig. 3' and 'Scheme 1', the reader is referred to the web version of this article.

a gradually blue-shift from 415 to 402 nm (Fig. 3d). Considering their different structures and UV–Vis adsorption behaviors, the variation of the photoluminescence emission should be owing to the distinct particle sizes and graphitic degrees of the CDs from the gradually increased temperatures [36,37]. The cystine produced CDs also had an obvious excitation-dependent photoluminescence behaviors (Figs. 3e and S5). Corresponding to the observation in the UV–Vis spectra, the PL emission spectra of CDs-Cys-200 is different from those of CDs-Cys-220 and CDs-Cys-250. The maximum excitation of CDs-Cys-200 is located at 330 nm, while the other two CDs show the similar maximum excitation at around 320 nm, which is similar to the phenomena of those CDs from histidine.

Although derived from different amino acids, the QYs of the CDs exhibit a similar decreasing trend with the elevated hydrothermal temperatures (Table 1), which should be owing to the reduced sizes and the loss of heteroatoms for the CDs at high temperature [36,37]. Moreover, these amino acid produced CDs have very high PL stability. Even after being kept for 1 year in air at room temperature, the CDs still exhibit a transparent appearance and strong blue color under UV light (365 nm; insert in Scheme 1), which offers another advantage for their future applications.

4. Conclusion

In this work, we developed an environmental-friendly one-pot approach towards photoluminescent CDs by hydrothermal carbonization of different amino acids. It is found that both the structure of starting materials and the hydrothermal temperature can effectively influence the morphology and photophysical properties of the resulting CDs such as graphitic degrees, the adsorption/emission peaks and the quantum yields. Combining their favorable optical properties and low cytotoxicity, these CDs hold promise of various applications for new type fluorescence marker, bio-sensors, drug delivery, and bio-imaging of tissues.

Acknowledgments

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.jcis.2014.10.030.

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